

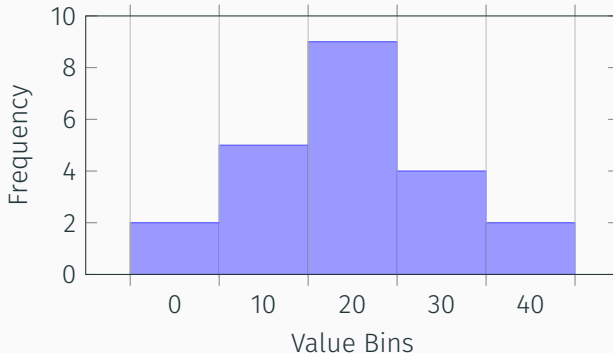
Data Visualization Foundations

Introduction to Data Science

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Understanding Distributions: Histograms

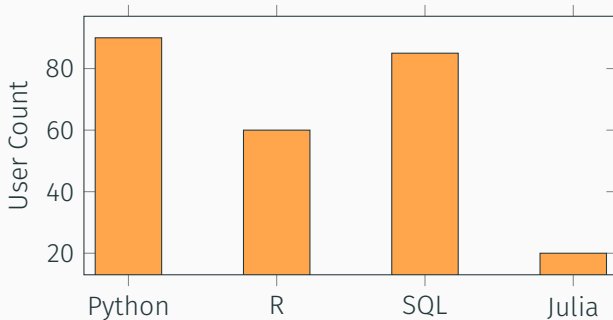
A **Histogram** groups continuous numerical data into bins to reveal the "shape" of the data.



- **No Gaps:** Indicates continuity on the X-axis.
- **Bins:** Selecting bin size is critical for detail vs. noise.

Comparing Categories: Bar Charts

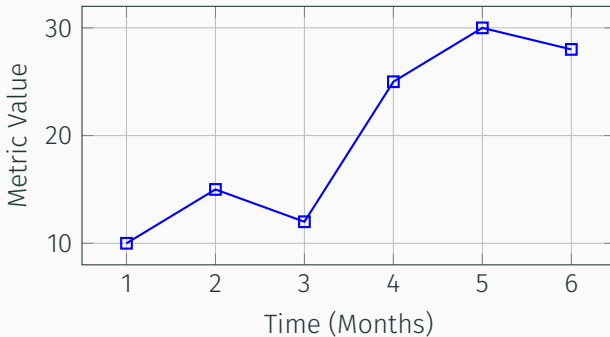
Bar Charts are used for discrete categories. Unlike histograms, the order is often arbitrary or sorted by value.



Comparison: Gaps between bars signal distinct categories.

Temporal Analysis: Line Graphs

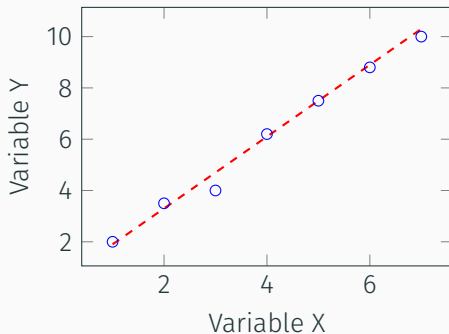
Line graphs connect data points to highlight the **rate of change** and trends over time.



Key Insight: Best for identifying seasonality and momentum.

Analyzing Relationships: Scatter Plots

Scatter plots examine the **correlation** between two numerical variables.

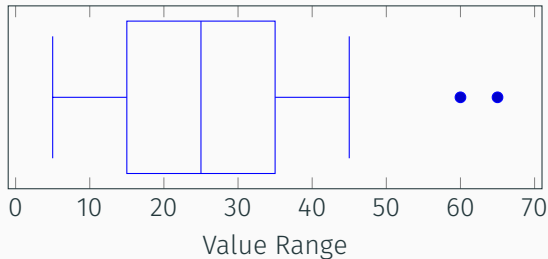


Direction: Positive, Negative, or Zero correlation.

Strength: How closely do points cluster around the trend line?

Statistical Summaries: Boxplots

The **Boxplot** visualizes the 5-number summary and flags outliers.



- **IQR:** The box contains the middle 50% of data.
- **Whiskers:** Show variability outside the upper and lower quartiles.
- **Dots:** Represent data points that are statistically "extreme."

Strategic Interpretation

When analyzing a visual, ask the following diagnostic questions:

Observation	Diagnostic Question
High Skew	Is the data naturally bounded (e.g., income)?
Outliers	Is this a measurement error or a rare event?
Non-linear Trend	Should we use a more complex model?
High Variance	Is our sample size sufficient?

Summary: Visualization is the first step in identifying data quality issues and hidden patterns before any formal modeling begins.